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Sardar Patel University
M. Sc. Integrated Biotechnology 7th Semester examination (IGMBT)
PS07EIBM24 - (Clinical Biochemistry)
Thursday, 7th January 2021
10:00 A.M. to 12:00 P.M.

Total marks: 70

Q.1 (A) Multiple choice questions: (08)

- 1 Some common causes of pleural effusions include:
 - a) Jaundice
 - b) Myocardial infraction
 - c) Rheumatoid arthritis
 - d) none of the above
- 2 Sodium citrate is an _____.
 - a) Preservative
 - b) coagulant
 - c) Anti-coagulant
 - d) none of the above
- 3 Normal ratio of bicarbonate buffer in plasma is _____.
 - a) 20:1
 - b) 4:1
 - c) 1:4
 - d) 16:1
- 4 Which of the following biochemical parameter is true in a case of Respiratory acidosis in uncompensated phase?
 - a) Disproportionate increase in $[H_2CO_3] \uparrow$
 - b) Increase in ratio of $[HCO_3^-] : [H_2CO_3]$
 - c) Decrease in total CO_2 content $\downarrow$
 - d) Decrease in $[HCO_3^-] \downarrow$
- 5 When hemoglobin is bound to O_2 , it is called _____.
 - a) Deoxyhemoglobin
 - b) Both a and d
 - c) Methemoglobin.
 - d) Oxyhemoglobin.
- 6 Vandenbergh test is used to detect
 - a) Fatty liver
 - b) Jaundice
 - c) Alcoholism
 - d) None of the above
- 7 Which hormone is responsible for obesity?
 - a) Insulin
 - b) epinephrine
 - c) leptin
 - d) All of the above
- 8 Thyroid status most accurately be mirrored by
 - a) Serum TBG concentration
 - b) Serum T4 level
 - c) T3 uptake
 - d) Free thyroxine index

Q.1 (B) Do as Directed (16)

- 1 Midstream clean catch specimen type of urine collection is strongly recommended for microbial culture for Pediatric patients. (True/False)
- 2 Ascites is a gastroenterological term for an accumulation of fluid in the Brain

[1]

[P.T.O.]

cavity (True/False)

- 3 The lymph is formed when the interstitial fluid is collected through lymph capillaries. (True/False)
- 4 Encephalitis is an inflammation due to Allergy in lungs. (True/False)
- 5 Metabolic Alkalosis Also called as primary alkali excess. (True/False)
- 6 _____ hormone shows antidiuretic effect and provides facultative reabsorption of water from distal convoluted tubule, henle's loop and collecting duct.
- 7 Na^+ , K^+ are consider as aprotes. (True/False)
- 8 Freshly synthesised chylomicrons contain _____.
- 9 IF the Genetic defect is *Homozygous*, the patient will have "trait". (True/False)
- 10 Unconjugated bilirubin is highly lipid soluble. (True/False)
- 11 *Decreased serum GOT level in pregnancy* and cirrhosis is due to enzyme inhibition. (True/False)
- 12 L-stercobilinogen is formed in _____.
- 13 BSP Retention test is consider in thyroid function test. (True/False)
- 14 Ketone-bodies are present in high concentration in uncontrolled Diabetes mellitus. (True/False).
- 15 Obesity is defined as excessive or abnormal fat accumulation in the body which result in overweigh to body. (True/False)
- 16 A single thyroid function test is sufficient for diagnostic it accurately. (True/False)

Q.2

Answer the following (Any Seven)

1. Enlist types of urine collection. (14)
2. What is cerebrospinal fluid?
3. What is volatile and nonvolatile acid? Enlist volatile and nonvolatile acid produced in the human body
4. What is respiratory alkalosis?
5. What are the sources of cell derived enzymes? Explain in brief.
6. Give the note on different forms of hemoglobin.
7. List out functions of liver.
8. List out different pathogenesis of atherosclerosis.
9. Write about type of obesity on the basis of energy imbalance.

Q.3

Write a detail account on pleural fluid.

(08)

OR

Q.3

Write detail note on collection, preservation and transportation of blood and Urine. (08)

Q.4

Write detail account on respiratory acidosis.

(08)

OR

Q.4

Write note on physiological regulation of electrolyte balance.

(08)

- Q.5 Discuss the metabolism of hemoglobin in detail. (08)
OR
Q.5 Write note on serum enzymes used for diagnosis of heart diseases. (08)
- Q.6 Classify liver function test. Explain test based on excretory functions of liver in detail. (08)
OR
Q.6 Write brief account on Diabetes (08)

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